

Startup Performance of UAV BLDC Motor Drives using Soft-Start Technique Under Open-Loop Control

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Introduction: Brushless DC (BLDC) motors are extensively used in UAV propulsion due to their high efficiency and favorable torque characteristics [1], [2]. The start-up behavior strongly influences battery loading, ESC durability, and flight stability. In normal open-loop start, sudden PWM causes high inrush and stress [3], while Soft-Start gradually ramps it to reduce electrical and mechanical transients [4]. This work evaluates both startup methods to quantify how much soft-start improves performance in UAV systems where battery capacity and ESC protection is important. Open-loop control is used in the startup because the motor is stationary at the beginning and produces no back-EMF for feedback, so a PWM ramp is needed to get it moving.

Methodology: The experimental setup uses an Arduino-based PWM signal to drive a BLDC motor through an ESC, with current and speed measured to compare normal start and soft-start behavior.

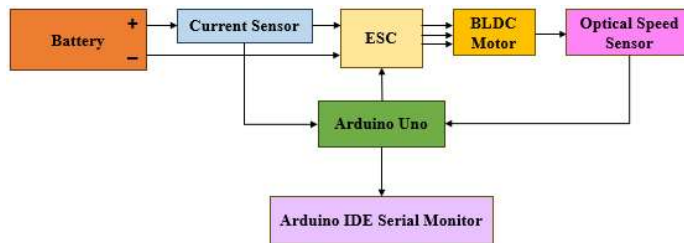


Fig. 1: Experimental setup block diagram.

Results: The Arduino-based setup measures current and speed for both approaches, showing that Soft-Start reduces peak current, peak power, and startup energy by nearly 24%, 29%, and 29%, respectively.

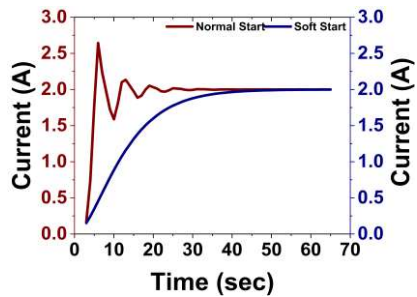


Fig. 2: Current

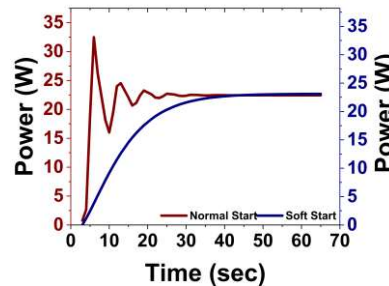


Fig. 3: Power response during motor startup.

response during motor startup.

Conclusion: The findings confirm that Soft-Start provides a clear operational advantage by lowering electrical stress and supporting better long-term reliability of UAV propulsion systems.

Keywords: BLDC motor, UAV propulsion, Inrush Current, Normal Start, and the Soft Start.

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